IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

EEE PART IV: MEng and ACGI

POWER SYSTEM ECONOMICS

Monday, 29 April 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Question 1

(a) Describe the significance of the concept of system marginal price and explain why it varies with time in the case of electricity.

[4]

(b) The electricity demand Q [MWh] in a given period is a function of price π [£/MWh] and is given by the following expression: $\pi = -0.1Q + 92$ [£/MWh]. The marginal cost of production in the period under consideration is: $MC = 0.15Q + 15$ [£/MWh]

(i) Calculate the level of consumption, price, consumer surplus, producer surplus, and the total social welfare at the equilibrium.

[4]

(ii) At the price $\pi = 65$ [£/MWh], calculate the consumption, consumer surplus, producer surplus, and the total social welfare at the equilibrium. How do these compare with the equilibrium calculated in (i)?

[4]

(c) Two generators are available to supply a demand of $D = 600$ [MW]. The cost of generating power $C_i(P_i)$ corresponding to each of the generators i is:

$$C_1(P_1) = 300 + 8.05 * P_1 + 0.2 * P_1^2 \ [\frac{£}{h}]$$

$$C_2(P_2) = 565 + 8.45 * P_2 + 0.3 * P_2^2 \ [\frac{£}{h}]$$

Calculate the optimal production of each generator, electricity price and profits of each generator.

[4]

(d) Assume that the maximum output limit of each generator is:

$$P_1^{max} = 350 \text{ [MW]}$$

$$P_2^{max} = 300 \text{ [MW]}$$

Determine the optimal production of each generator, electricity price and the profits of each generator. Based on the results, briefly explain the impact of considering these maximum output limits on the total cost of operating the system and the profits of the two generators.

[4]

Question 2

(a) Explain briefly why electricity markets may lend themselves to be imperfect competitive markets.

[4]

(b) Consider a market for electrical energy that is supplied by two generating companies whose cost functions are:

$$C_A = 75 * P_A \left[\frac{£}{h}\right]$$

$$C_B = 62 * P_B \left[\frac{£}{h}\right]$$

where P_A and P_B are production levels of the two generating companies respectively. The inverse demand function for this market is estimated to be:

$$\pi = 250 - 2D \left[\frac{£}{MWh}\right]$$

where π and D are the electricity price and demand respectively.

(i) If we assume a Bertrand model for the competition in this market, calculate the production of each of the companies, the resulting market price and profits made by both companies.

[2]

(ii) If we assume a Cournot model of competition, form a table to identify the Nash Equilibria (NE) of this market, i.e. market price, demand quantity and profit of each company for different levels of production for each of the companies. Consider a range of production levels for company A of 20MW, 25MW, 30MW and 35MW, and for company B of 25MW, 30MW, 35MW and 40MW.

[8]

(c) It is observed in (b)-ii that one of the cells of the derived Cournot table involves higher profits, for both competing companies, than the one corresponding to the NE. Explain why this cell does not express the equilibrium state of the market.

[2]

(d) State the optimality conditions and evaluate the exact NE of this market. At the NE, calculate the market price, demand quantity and profit of each company.

[4]

Question 3

Consider the three-bus power system shown in Figure Q3. Demands at buses 2 and 3 and line reactance values are indicated in the figure.

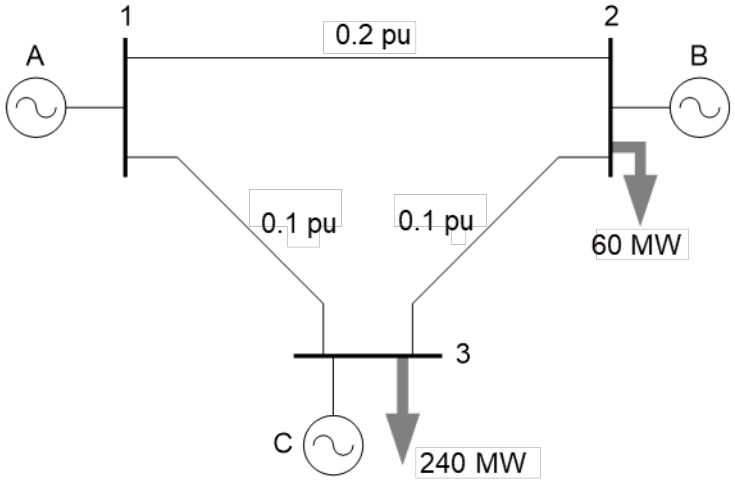


Figure Q3. A simple three-bus system

Assume that generating units A, B and C have the following marginal production costs:

$$MC_A = 12 + 0.1P_A \text{ [£/MWh]}$$
$$MC_B = 18 + 0.2P_B \text{ [£/MWh]}$$
$$MC_C = 45 \text{ [£/MWh]}$$

(a) Calculate the optimal unconstrained dispatch for these conditions, and the nodal prices. What is the hourly cost of this dispatch?

[3]

(b) Calculate the power flows in each line of the network if this dispatch was implemented.

[4]

(c) Assume that the flow in line 1-3 is limited to 160 MW for security reasons. How should the generators be re-dispatched in order to supply the demand at minimum cost without violating line 1-3 flow limit?

[5]

(d) Calculate the hourly cost of this constrained dispatch and the hourly cost of security.

[3]

(e) Calculate the marginal cost of energy at each node after the constraint on the flow on line 1-3 is taken into consideration. What is the total congestion surplus?

[5]

Question 4

(a) List and explain key characteristics of electricity transmission as a stand-alone business.

[6]

(b) Consider a two-area system, with demand in Area A D_A=1500 [MW] while demand in Area B is D_B= 400[MW]. Generator A is in Area A and generator B is located in Area B, with their respective cost functions:

$$C_A(P_A) = 1000 + 10 * P_A + 0.01 * P_A^2 \ [\frac{£}{h}]$$

$$C_B(P_B) = 500 + 5 * P_B + 0.005 * P_B^2 \ [\frac{£}{h}]$$

Determine the optimal generation dispatch, locational marginal prices, total cost of operating the system, cost of constraints and congestion surplus for:

(i) The case where the capacity of the transmission link between the two areas is not binding.

[3]

(ii) The case where no transmission link exists between the two areas.

[3]

(c) If the annuitized cost of building the transmission line is given by $C_{inv} = k * L * F$, where F is the capacity of the line, $L = 750km$ (length of the line) and $k = 120 \ [£/(MW.km.year)]$, calculate:

(i) Transmission demand function;

[1]

(ii) Transmission supply function;

[1]

(iii) Congestion surplus as a function of F.

[2]

(d) Determine the optimal capacity that should be built if

(i) Transmission is a regulated activity;

[1.5]

(ii) Transmission is operated as a merchant company.

[2.5]